ETHICS AND RESEARCH

Dr. Rabab Ali Abumalloh

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

Structure

RESEARCH PAPER

1 Introduction

Explains the research topic, why it is important, and what the study aims to investigate.

2 Literature Review

Summarizes and compares previous studies to show what is already known and what is missing.

3 Method

Describes how the study was conducted, including participants, tools, and data collection procedures.

4 Results

Presents the findings of the study using numbers, tables, or themes without interpretation.

5 Conclusion

Interprets the results, links them to previous research, explains implications, and summarizes the study.

TIMELINE

1 Topic Selection

Choose a clear research topic and define the main problem to study.

2 Literature Review

Read academic papers and identify key theories, findings, and research gaps.

3 Research Design

Plan the methodology, select participants, and prepare data collection tools.

4 Data Collection

Collect data using surveys, experiments, interviews, or observations.

5 Data Analysis

Analyse the data using statistical tools or qualitative methods.

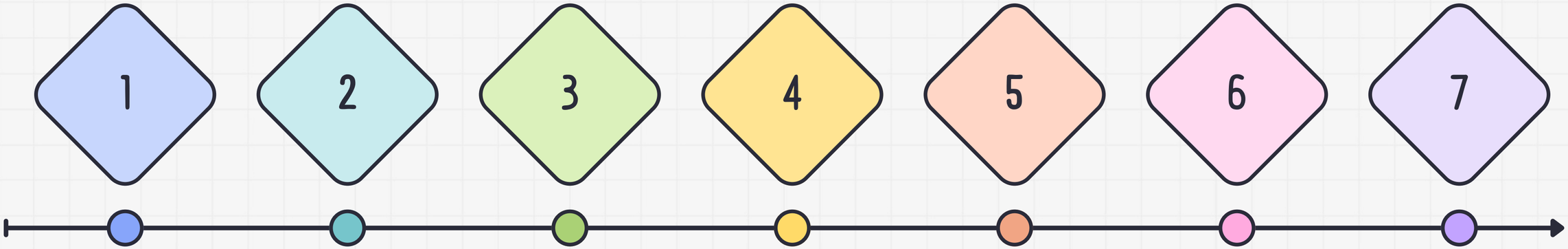
6 Writing Draft

Write the introduction, literature review, methods, and results sections.

7 Submission

Edit, proofread, format the paper, and submit the final version.

RESEARCH TOPIC SELECTION



1 Interest Area

Choose a subject you are curious about and motivated to study.

First milestone

2 Initial Reading

Read basic articles and reports to understand the topic area.

Second milestone

3 Identify Problems

Find real issues, challenges, or unanswered questions.

Third milestone

4 Narrow the Focus

Make the topic specific and manageable.

Fourth milestone

5 Feasibility Check

Check if the topic is practical (time, data, access, tools).

Fifth milestone

6 Research Question

Turn the topic into a clear research question.

Sixth milestone

7 Final Topic Approval

Confirm the topic

Seventh milestone

ABSTRACT WRITING CHRONOLOGY

Step 1
Understand the Study
Clearly identify the purpose, topic, and research problem.

Step 2
Identify Methods
Decide how the study was conducted (survey, experiment, interview, etc.).

Step 3
Summarise Key Results
Select the most important findings only (no details).

Step 4
State Main Conclusion
Explain what the results mean in one clear sentence.

Step 5
Highlight Contribution
Show why the study is useful or important.

Step 6
Check Word Limit
Edit to fit the required length (usually 150–250 words).

Step 7
Final Review
Check clarity, grammar, and academic tone.

RESEARCH METHODS

Co-Design

Work with users to co-create ideas, systems, or solutions.

2025

Survey

Collect quantitative data using structured questionnaires.

2026

SWOT Analysis

Analyse strengths, weaknesses, opportunities, and threats.

2027

Experiment

Test cause-and-effect relationships under controlled conditions.

2028

Interviews

Conduct interviews to explore experiences and perceptions.

2029

Delphi Method

Collect expert opinions through multiple anonymous rounds to reach consensus.

2030

ETHICS

1

Integrity

Be honest in designing, conducting, and reporting research.

2

Beneficence

Ensure the study benefits society and avoids harm.

3

Informed Consent

Participants must understand the study and agree voluntarily.

4

Data Protection

Store and manage data securely and legally.

5

Confidentiality

Protect participants' identity and personal data.

6

Accountability

Take responsibility for ethical conduct and corrections.

7

Transparency

Clearly report methods, funding, and conflicts of interest.

AI USAGE IN RESEARCH

Idea Generation

Use AI to brainstorm topics, refine research questions, and outline papers.

1

Literature Support

Use AI to summarize papers and identify key themes.

2

Instrument Design

Use AI to help design surveys, interview questions, and scenarios.

3

Data Analysis

Use AI to support data coding, pattern detection, and visualization.

4

Writing Support

Use AI to improve language, structure, and academic style.

5

Ethical Use

Use AI responsibly with transparency and proper citation.

6